Mingyang (Yang) Liu

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Placement administrator: Ms Kim Hart k.hart@imperial.ac.uk

Education

Imperial College London, Imperial Business School, Department of Finance — Aug 2019 – Aug 2026
PhD in Finance, conferred August 2026
MRes in Finance, with Distinction, 2020

Columbia University, Graduate School of Arts and Sciences — Aug 2017 – Feb 2019
M.A. in Economics (Financial Econometrics; STEM-designated programme)
Master's thesis: Diffusion Index Forecasting: A Robust Estimator for Common Shock, supervised by Prof. Jushan Bai

University of Michigan, Ann Arbor — Aug 2014 – Apr 2017
B.S., double major in Economics and Mathematics (Mathematics of Finance and Risk Management track)

Fields

Empirical asset pricing | Machine learning and big data in finance | Financial econometrics
Portfolio choice and risk management

References

Professor Paolo Zaffaroni
Professor of Financial Econometrics
Imperial Business School
Imperial College London
p.zaffaroni@imperial.ac.uk
PhD supervisor (main)

Professor Pasquale Della Corte
Professor of Finance
Imperial Business School
Imperial College London
p.dellacorte@imperial.ac.uk
PhD supervisor

Professor Robert Kosowski
Professor of Finance
Co-Director, Imperial Centre of Excellence in Quantitative Finance
Imperial Business School, Imperial College London
r.kosowski@imperial.ac.uk
Host, visiting researcher appointment

Fellowships & Awards

Fully funded doctoral scholarship (Graduate Teaching Assistant Scholarship), Imperial Business School — 2019 – 2025
MRes in Finance with Distinction, Imperial College London — 2020
Nominated by students for the Excellence in Teaching Award, Imperial Business School — 2023, 2024
University Honors, University of Michigan — 2014 – 2016

Job Market Paper

The Ruler Matters: Factor Exposures, Attribution, and Portfolio Risk
Working paper, October 2026 | PDF | Paper page

Firm characteristics parameterize conditional factor loadings in IPCA. A larger characteristic library adds new directions while also including directions that overlap with those already represented. We use the cross-sectional distribution of firm characteristics as a ruler to evaluate fixed-dimensional IPCA loading spaces. Holding the number of factors fixed, expanding from 36 to 153 characteristics can improve in-sample fit while representing momentum and size positions less fully. Reallocating shared factor exposures changes their attribution without changing joint fitted returns. Holding model coefficients fixed and the characteristic target constant, changing the exposure reference changes stock holdings and trading demands, as well as subsequent return risk. The findings show that factor models should be evaluated by the economic exposures they represent and the portfolio decisions those exposures support, alongside return fit.

Seminar and conference presentations: University of Cambridge, Faculty of Economics, 4 November 2026 (invited; scheduled); EPFL, Lausanne, 16 June 2026 (invited seminar; precursor presented as No-Arbitrage Factor Geometry)

Working Papers

Same Forecasts, Different Portfolios: Factor Normalization and Regularization

Working paper, October 2026 | PDF | Paper page

Portfolio Sharpe ratios from statistical factor models can reflect allocation choices even when return forecasts coincide. Adding a ridge adjustment to factor covariances penalizes factor exposures in units set by normalization. We derive the transformation that preserves portfolio positions across equivalent model representations. Two separately estimated IPCA normalizations deliver nearly identical forecasts but gross Sharpe ratios of 0.69 and 1.07 under the same numerical penalty; transforming the penalty consistently aligns their portfolios. Using the full fitted mean, including characteristic-associated intercepts, produces a portfolio with a gross Sharpe ratio of 2.51 and little sensitivity to the penalty.

Characteristic-Space Metrics in Factor Models: Identification and Inference

SSRN preprint, September 2026 | DOI 10.2139/ssrn.7445120 | PDF | Online supplement

Factor models with identical fit can credit returns differently to latent factors, observed factors and an intercept; fit cannot choose the split, but the inner product chosen for characteristic space does. The paper derives rotation-invariant inference that carries the Gram metric's estimation error, and drift-robust confidence intervals near an intercept cap (exposure budget).

A Geometric Framework for Identification in Characteristic-Based Factor Models

SSRN preprint, March 2026 (preliminary version) | DOI 10.2139/ssrn.7013178 | PDF | grows out of the PhD thesis; its inference theory is developed further in Characteristic-Space Metrics

Using a metric taken from the long-run cross-sectional distribution of firm characteristics, the paper derives NA-IPCA, which nests IPCA, caps how much the intercept may carry and admits rotation-invariant inference. In the headline calibrated simulation, NA-IPCA recovers the latent pricing span and an intercept-free IPCA benchmark does not; that design does not isolate the metric.

Interpreting Estimated Pricing Errors: Evidence from Characteristic-Based Return Forecasts

Working paper, September 2026 | PDF | Paper page

What does a smaller estimated pricing error (alpha) tell a model user? On 153 U.S. characteristics, crossing the expected-return estimates and factor spaces of one model's equal- and value-weighted fits shows that their gap in fitted intercept magnitudes mainly reflects different estimated means; residual diagnostics add little to forecast selection (exploratory evidence).

Research in Progress

Exposure or Intercept? Identifying Corporate Bond Premia with No-Arbitrage IPCA, with Paolo Zaffaroni

In preparation

Characteristics such as rating, duration and liquidity predict corporate bond returns, but whether this reflects factor exposure or an intercept the factors leave unexplained remains unsettled. Because these characteristics overlap heavily and rest on sparse transaction prices, we ask how much of this split the data identify and how much modelling conventions decide. We adapt No-Arbitrage IPCA (NA-IPCA) to corporate bonds: characteristics instrument latent and observed factor loadings and an intercept, loadings are normalised under the characteristics' Gram metric, and a no-arbitrage cap limits how much the intercept may carry. Zero and unrestricted intercepts, compared in earlier IPCA work on bonds, are the cap's two ends. We estimate the model on the public Open Source Bond Asset Pricing panel of TRACE excess returns over duration-matched Treasuries, following the replication literature's data protocol. Out of sample, we evaluate how the split, and whether the cap binds, depend on the metric, the cap and the observed factors.

Ongoing research collaboration with Prof. Semyon Malamud (EPFL and Swiss Finance Institute) on portfolio construction from linear out-of-sample return forecasts. The project complements his work on nonlinear return prediction and began after a seminar presented at EPFL in June 2026 at his invitation. Jul 2026 – present

PhD Thesis

No-Arbitrage IPCA: A Framework for Cross-Sectional Asset Pricing

Imperial College London, 2026 | supervisors: Prof. Paolo Zaffaroni (main) and Prof. Pasquale Della Corte | examined March 2026; degree conferred August 2026 | available on request

Extends IPCA, via the empirical characteristic metric, to nested QZ-IPCA and NA-IPCA models separating latent, observable (Fama–French five factors plus momentum) and residual-pricing components under an intercept cap; gives identification results, an asymptotic linear expansion and projected alternating least squares; simulations and a Jensen–Kelly–Pedersen U.S. panel show near-identical fit coexisting with different identified pricing directions.

Teaching

Financial Statistics (MSc Risk Management & Financial Engineering), Imperial Business School; Graduate Teaching Assistant — 2023, 2024

Student rating 4.8 out of 5; nominated by students for the Excellence in Teaching Award in both years. Teaching notes: Financial Statistics (PDF, 96 pages, shared for academic use).

Affiliations & Research Positions

Visiting Researcher, Imperial Centre of Excellence in Quantitative Finance, Department of Finance, Imperial Business School, Imperial College London — Jun 2026 – present

Host: Prof. Robert Kosowski, Co-Director of the Centre. Research agenda for the visit: economically constrained machine learning for cross-sectional asset pricing and systematic investment.

Research Assistant (macro forecasting), Department of Economics, Columbia University — May 2018 – May 2019

Research Assistant (derivatives pricing), Department of Mathematics, University of Michigan — Jun 2016 – Jan 2017

Service & Leadership

Adviser; former Chair of the Executive Committee and Advisory Committee, Imperial Student Investment Fund — May 2024 – present

Helped the Head of the Department of Finance set up a lab for the fund, linking teaching, research and industry practice; helped build its Python multi-asset research and trading platform; helped bring in the Trading and Risk Management programme of Qube Research & Technologies (QRT) and portfolio-optimisation training on LSEG Workspace; mentored the fund's student teams in their first QRT Quantitative Competition (2026; four teams reached the final, two placed second and third).

Other Activities

Independent Quantitative Research & AI/ML Consultant: project-based advisory engagements for emerging systematic investment teams (research infrastructure, data and feature engineering, Python research workflows), in a personal capacity, separate from the Imperial appointment.

Languages

Chinese (Mandarin, native); English (fluent; working language)

Software

Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, PyTorch), MATLAB, R, SQL, Git. Author of `naipca`, a Python toolkit for IPCA, QZ-IPCA and NA-IPCA with a companion metric-diagnostics package (available to academic users on request).

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